

AIML Lab Report: Experiment 8-9

DBSCAN Clustering with PCA on Air Quality Data

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1 Objective

The goal of this experiment (Lab 8-9) was to apply the DBSCAN clustering algorithm to identify density-based clusters within the `city_day.csv` air quality dataset. To aid in visualizing the high-dimensional data and potential cluster structures, Principal Component Analysis (PCA) was utilized for dimensionality reduction prior to visualization [1].

2 Methodology

2.1 Step 1: Importing Libraries

The experiment started by importing the necessary Python libraries for data manipulation, numerical operations, visualization, data preprocessing, clustering, and dimensionality reduction [1].

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from sklearn.cluster import DBSCAN
6 from sklearn.preprocessing import StandardScaler
7 from sklearn.decomposition import PCA
```

Listing 1: Library Imports

2.2 Step 2: Data Loading and Preprocessing

The following preprocessing steps were performed on the `city_day.csv` dataset [1]:

1. **Loading Data:** The dataset was loaded into a pandas DataFrame. An initial check with `df.info()` revealed the presence of missing values and mixed data types (objects and float64).
2. **Handling Missing Values:** Rows containing any NA values were removed using `df.dropna(inplace=True)`. This significantly reduced the dataset size to 6236 rows, ensuring only complete records were used for clustering.
3. **Feature Selection:** Non-numeric columns ('City', 'Date', 'AQI_Bucket') were dropped to isolate the 13 numerical features (pollutant levels and AQI value) relevant for clustering based on measurement similarity.
4. **Feature Scaling:** The remaining numerical features were standardized using `StandardScaler` from scikit-learn. This scaling transforms the data to have zero mean and unit variance, which is important for distance-based algorithms like PCA and DBSCAN, preventing features with larger ranges from disproportionately influencing the results.

```
1 # df = pd.read_csv('path/to/city_day.csv') % Replace path
2 # df.info() # Initial check
3 df.dropna(inplace = True)
4 df.drop(['City' , 'Date' , 'AQI_Bucket'] , axis =1 , inplace = True)
5 scaler=StandardScaler()
6 scaled=scaler.fit_transform(df)
```

Listing 2: Data Preprocessing Steps

2.3 Step 3: Dimensionality Reduction (PCA)

Principal Component Analysis (PCA) was applied to the scaled numerical data to reduce the dimensionality from 13 features down to 2 principal components, primarily for visualization purposes [1]. These two components capture the directions of maximum variance in the data.

```
1 pca = PCA(n_components=2)
2 pca_result = pca.fit_transform(scaled)
3 # The pca_result numpy array now holds the 2D representation
```

Listing 3: PCA Application

2.4 Step 4: DBSCAN Clustering

The DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm was implemented using scikit-learn [1].

- **Initialization:** DBSCAN was initialized with the standard parameters: ‘eps=0.5’ (the maximum distance between two samples for one to be considered as in the neighborhood of the other) and ‘min_samples = 5’ (the number of samples in a neighborhood for a point to be considered as a core point). This means clusters were identified based on density in the original 13-dimensional scaled feature space.
- **Prediction:** Cluster labels were assigned using ‘fit_predict()’. Points not belonging to any cluster (i.e., in low-density regions) were labeled as noise (-1).
- **Storing Labels:** The resulting cluster labels were added as a new column to the data frame.

```
1 dbscan = DBSCAN(eps=0.5, min_samples=5).fit(scaled) # Fit on scaled
  data
2 clusters = dbscan.fit_predict(scaled) # Predict on scaled data
3 df['DBSCAN_Cluster'] = clusters # Add labels to dataframe
```

Listing 4: DBSCAN Application

3 Results

3.1 Identified Clusters

The DBSCAN algorithm partitioned the 6236 data points into several clusters and identified a large number of noise points. The distribution of points per cluster label was [1]:

- DBSCAN_Cluster
- | | | |
|-----|--|---------|
| -1 | 4588 | (Noise) |
| 0 | 1137 | |
| 7 | 106 | |
| 10 | 55 | |
| 12 | 50 | |
| 11 | 21 | |
| 15 | 17 | |
| 39 | 16 | |
| 29 | 14 | |
| 8 | 14 | |
| 9 | 13 | |
| 13 | 11 | |
| 24 | 11 | |
| 1 | 11 | |
| ... | (other smaller clusters listed in notebook output) ... | |
| 14 | 3 | |

With the parameters ‘ $\text{eps}=0.5$ ’ and ‘ $\text{min_samples} = 5$ ’, the majority of the data (4588 points, approx. 73.6%) was classified as noise.

3.2 Visualization

The clusters were visualized using a scatter plot based on the 2D PCA results (Figure 1). The x and y axes represent the first and second principal components, respectively. Each point was colored according to the cluster label assigned by DBSCAN [1].

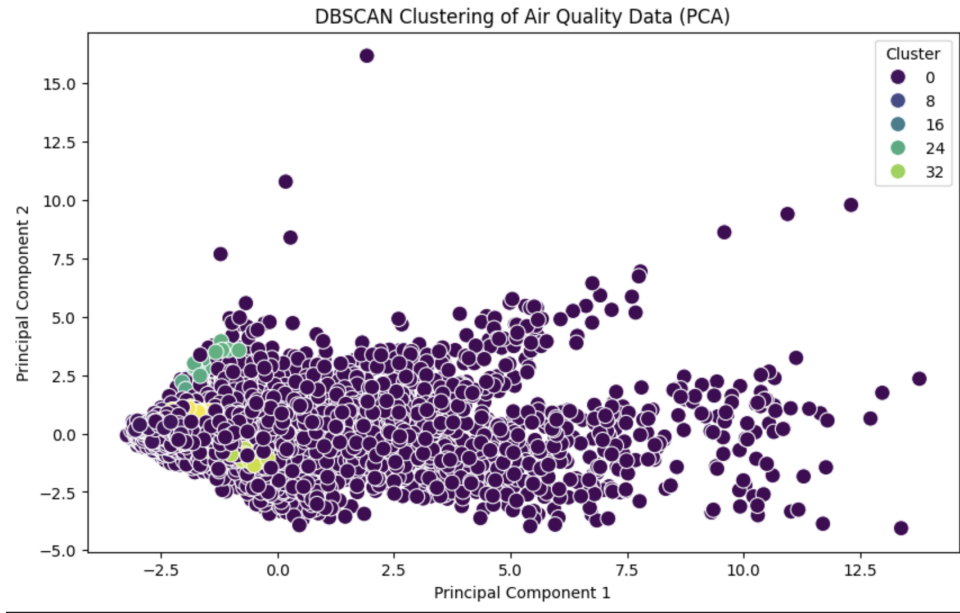


Figure 1: Scatter plot of DBSCAN clustering results projected onto the first two Principal Components. Colors indicate cluster assignment (noise points typically shown differently).

This visualization (Figure 1) helps to assess the spatial distribution of the groups in the dimensions that capture the most variance. Although the clusters were formed in the original scaled space, plotting them on the PCA axes provides insight into their structure and separation as represented by the principal components.

4 Discussion and Conclusion

This experiment successfully applied DBSCAN clustering to a pre-processed air quality data set, using PCA primarily as a tool to visualize high-dimensional cluster results in 2D [1].

The DBSCAN algorithm, using default parameters, identified a clear dominant cluster (cluster 0) and numerous smaller clusters, characteristic of its density-based approach, which can find clusters of arbitrary shapes. However, a very large proportion of the data was labeled as noise (-1). This outcome could indicate several possibilities:

- The data inherently contains many sparse regions or outliers that do not meet the density criteria set by ‘ $\text{eps}=0.5$ ’. The PCA visualization proved useful for observing the resulting cluster assignments in a reduced space, showing the relative positions and spread of the identified groups and noise points based on the principal components.

In conclusion, the combination of PCA and DBSCAN provided insight into the density structure of the air quality data. The significant number of noise points suggests that further exploration with different DBSCAN parameters (perhaps guided by methods like k-distance plots for ‘ eps ’

estimation) or alternative clustering algorithms might be beneficial for a more comprehensive understanding of the data grouping patterns.

References

[1] Source: 'Lab8-9.ipynb' notebook content and outputs.